

Editor
Remote Sensing

Dear Editor,

Please find our substantially revised manuscript, “Statistical Evidence for a DQN-Family Advantage in a Resource-Coupled Satellite-Ground Scheduling Benchmark.” We thank you and the three reviewers for the constructive assessment. The title has been retained, and the clean and highlighted manuscripts contain identical scientific content.

The revision adds discount-factor and training-length experiments for all six DQN objectives, complete six-objective learning curves, an eight-step greedy rollout, and a matched single-thread CPU decision-time comparison. It also narrows every principal claim to the synthetic, clipped-resource, soft-constraint benchmark; recasts the fixed- λ component comparisons as descriptive; and expands the discussion of hard constraints, MILP, dynamic programming, PPO, SAC, forecast information asymmetry, structural model error, simulator provenance, and real-mission validation. All new learned-policy experiments use 20 prespecified model seeds and 20 shared held-out traces per seed. Inference uses 10,000 model-seed-block bootstrap resamples and separate Holm families.

For convenient editorial review, the complete reviewer comments and author responses are reproduced below verbatim from the three reviewer-specific response documents. Each entry also gives the revised manuscript excerpt and its exact location in the final clean manuscript PDF.

Response to Reviewer 1

Comment. *The abstract and conclusions repeatedly refer to “satellite-ground scheduling” without explicitly qualifying that the results apply only to a soft-constrained, clipped-resource benchmark. Real missions impose hard energy/thermal limits. Please reword the abstract and conclusion to clearly delimit the scope, and add a short paragraph in the Discussion assessing how hard constraints might affect the observed ordering.*

Response. We agree. The Abstract, Introduction, Discussion, Highlights, and Conclusions now identify the study as a synthetic benchmark with clipped, soft-constrained resources. We also explain that hard constraints change the feasible action set through action masking, task rejection, or protective shutdown and may therefore change the observed ordering.

Revised manuscript text.

These findings describe this synthetic soft-constraint benchmark only; hard energy and thermal limits can change the feasible action set and therefore the algorithm ordering.

Location in the revised manuscript PDF. p. 1, Abstract and Highlights; p. 3, Introduction; p. 23, Section 5.1; p. 24, Conclusions.

Comment. *The rationale for selecting exactly these six variants is not well motivated, and the auxiliary weight $\lambda = 0.3$ was calibrated only for the centred objective then reused for other auxiliary losses. This makes the component-wise comparisons (Tables 7–8) descriptive rather than fair hyperparameter-optimised contrasts. Please either run a separate grid search for each*

loss, or explicitly state that these comparisons are not intended to isolate the effect of centering/bootstrapping, and tone down the corresponding claims.

Response. We now motivate the six objectives as matched factual-only, full-action, centered, bootstrapped, and Double-DQN controls. We retained the prespecified $\lambda = 0.3$ and explicitly characterize Tables 7–8 as fixed-coefficient descriptive comparisons. The text no longer claims that these rows independently identify the effects of centering or bootstrapping.

Revised manuscript text.

Because the auxiliary losses differ in scale and error structure, these are fixed-coefficient implementation comparisons. They do not factorially isolate centering or bootstrapping.

Location in the revised manuscript PDF. p. 11, Section 3.4.1; pp. 17–18, Tables 7–8 and accompanying Results text; p. 23, Section 5.1.

Comment. *MPC-4 is the strongest non-DQN comparator, but the literature includes more advanced optimisation methods (e.g., MILP, dynamic programming). While the authors mention this in future work, the current comparison remains limited. I request at least a literature-based discussion of how DQN might fare against such methods, and, if feasible, add one additional lightweight optimisation baseline (e.g., a greedy look-ahead heuristic) for the nominal regime.*

Response. We added literature-based discussion of mixed-integer programming, dynamic programming, PPO, and SAC, and we state explicitly that the study does not establish universal DQN optimality. We also implemented an eight-step greedy rollout that enumerates the first action, completes the predicted rollout with one-step greedy actions, executes the first action, and replans. It receives the same perfect forecast as MPC. The validated nominal result and timing comparison are reported in the revised Results (pp. 20–21, Table 12) and reproducibility package.

Revised manuscript text.

The greedy rollout reduced mean cost relative to MPC-4 (88.79 versus 90.83) but remained above all six DQN means (77.66–78.40). Every paired DQN–greedy difference was negative, ranging from -11.12 to -10.39 ; all 95% bootstrap intervals excluded zero and the largest Holm-adjusted sign-test value was 0.000011.

Location in the revised manuscript PDF. pp. 2–3, Introduction and Related Work; pp. 12–13, Section 3.4.4; pp. 20–21, Section 4.6 and Table 12; p. 23, Section 5.2.

Comment. *The paper provides sensitivity on task parameters but not on the discount factor γ or training length. Since these directly affect value learning, please add a supplementary experiment or a sensitivity plot for $\gamma \in \{0.9, 0.95, 0.99\}$ on the nominal regime, and report learning curves for all six objectives (not just two) to confirm stable convergence.*

Response. We added a nominal experiment for all six objectives at $\gamma \in \{0.90, 0.95, 0.97, 0.99\}$ with 20 seeds and the same held-out traces. MPC-4 uses the corresponding discount. We also trained each objective to 900 episodes with evaluations at 300, 600, and 900 episodes while fixing epsilon decay at 13,440 interactions. The training figure now shows all six objectives. New inference families use separate Holm corrections. These additions appear on pp. 19–20 (Figs. 6–7).

Revised manuscript text.

At $\gamma = 0.95$, none of the six contrasts with $\gamma = 0.97$ was supported after Holm correction. Reducing γ to 0.90 increased mean cost for every objective by 2.63–4.57, with all adjusted sign-test values at or below 0.0057.

Location in the revised manuscript PDF. p. 14, Section 3.4.5; pp. 19–20, Sections 4.4–4.5 and Figures 6–7.

Comment. *The “counterfactual” preview is clearly defined, but the misspecification test only scales costs/deltas by 0.85/1.15; it does not test structural errors (e.g., wrong thermal dynamics). Please acknowledge this limitation and suggest how more realistic misspecification could be examined in future work.*

Response. We agree and now distinguish amplitude perturbation from structural error. The Discussion identifies incorrect thermal recurrence, actuator saturation, state-dependent contact loss, telemetry-calibrated alternate dynamics, and hardware-in-the-loop evaluation as future tests.

Revised manuscript text.

Preview misspecification was examined using uniform $\pm 15\%$ scaling, which tests amplitude error but not structural error such as an incorrect thermal recurrence, unmodeled actuator saturation, or state-dependent contact loss.

Location in the revised manuscript PDF. p. 22, Section 5; p. 23, Sections 5.1–5.2.

Response to Reviewer 2

Comment. *The current introduction mainly emphasizes that long-term value function learning outperforms myopic planning methods, but it fails to elaborate on the practical dilemmas in existing satellite-ground scheduling research and engineering practice. Specifically, it does not explain why traditional scheduling methods such as MPC and dynamic programming are difficult to deploy in practical scenarios, nor does it distinguish between hard constraints and soft constraints in real satellite-ground scheduling systems. All experiments in this paper are conducted based on a soft-constraint clipping resource simulator, where the over-limit of resources such as energy and heat is only penalized without causing hardware failures. Therefore, it is necessary to supplement relevant explanations in the introduction and limitation sections: real satellite systems are dominated by hard constraints, where resource overrun will directly lead to task failure or hardware protection shutdown. The conclusions of this paper are only applicable to the soft-constraint simulation environment and cannot be directly transferred to real on-orbit satellite systems. It is recommended to add a dedicated paragraph to clarify the core differences between soft constraints in simulation and hard constraints in real satellite systems.*

Response. We added a dedicated Introduction paragraph and expanded the scope and limitations. The revised manuscript explains that clipping and penalties allow an episode to continue after a resource overrun, whereas operational systems can mask actions, reject tasks, or enter protective shutdown. We therefore delimit every principal claim to the synthetic soft-constraint benchmark.

Revised manuscript text.

A hard constraint changes the feasible action set and can make a policy with a lower penalized cost infeasible. Consequently, the ordering observed below is a statement about the specified soft-constraint benchmark and cannot be transferred directly to an on-orbit controller.

Location in the revised manuscript PDF. pp. 2–3, Introduction; p. 23, Section 5.1; p. 24, Conclusions.

Comment. *The baseline algorithms adopted in this paper only include MPC-4, contextual bandit, simple threshold heuristic, and instantaneous cost minimization (argmin). Several representative mainstream algorithms in satellite edge scheduling are missing, including classic deep reinforcement learning (DRL) scheduling algorithms such as PPO and SAC, as well as traditional operation research baselines such as mixed integer programming (MIP) and rolling horizon heuristic scheduling. The current comparison only proves that the DQN family outperforms MPC-4, which is insufficient to demonstrate that DQN-based algorithms are the optimal algorithm set on the proposed benchmark. It is recommended to supplement the experimental results of at least 1–2 commonly used scheduling baselines (e.g., PPO) on this benchmark. If supplementary experiments cannot be completed due to computing resource limitations, the limitations of the current comparison should be clearly stated in the discussion section, and the potential performance of other alternative algorithms should be further discussed.*

Response. We added an eight-step greedy rolling-look-ahead baseline under the same perfect forecast used by MPC-4. We also added verified literature discussion of PPO, SAC, mixed-integer programming, and dynamic programming, and narrowed the claim to the implemented comparator set. The revised manuscript states that these alternatives may outperform DQN under different hard-feasibility, control-space, or global-look-ahead conditions. The new baseline is reported on pp. 20–21 (Table 12).

Revised manuscript text.

The greedy rollout reduced mean cost relative to MPC-4 (88.79 versus 90.83) but remained above all six DQN means (77.66–78.40). Every paired DQN–greedy difference was negative, ranging from −11.12 to −10.39; all 95% bootstrap intervals excluded zero and the largest Holm-adjusted sign-test value was 0.000011.

Location in the revised manuscript PDF. pp. 2–3, Related Work; pp. 20–21, Section 4.6 and Table 12; p. 23, Section 5.2.

Comment. *Section 3.4.1 clarifies that the auxiliary loss weight $\lambda = 0.3$ is only tuned for the centered full-action model, and all other DQN variants directly adopt this hyperparameter without independent optimization. Such experimental setting makes it impossible to decouple the individual contributions of the centralization mechanism, Bellman bootstrap strategy, and auxiliary loss coefficient to the overall performance. The results in Table 7 only reflect the combined effect of multiple factors and cannot explain the internal optimization mechanism of the algorithm. Supplementary ablation experiments are suggested to explore the independent effectiveness of each module and hyperparameter.*

Response. We agree with the identifiability concern. A separate grid search for every auxiliary objective is outside the present revision scope. We therefore recast these rows as fixed-coefficient descriptive comparisons, explain the scale mismatch among auxiliary losses, and remove causal component claims. The manuscript now states that factorial identification requires objective-specific tuning and controlled gradient normalization.

Revised manuscript text.

The uncentered full-action, immediate-advantage, and Double DQN auxiliary variants inherited $\lambda = 0.3$ without method-specific calibration. Because the auxiliary losses differ in scale and error structure, these are fixed-coefficient implementation comparisons.

Location in the revised manuscript PDF. p. 11, Section 3.4.1; pp. 17–18, Tables 7–8; p. 23, Sections 5.1–5.2.

Comment. *The MPC-H planner adopted in this paper possesses perfect forecast information of all future tasks and link states in the subsequent H steps, while the DQN policy can only observe the current system state without access to future task information. The inconsistent information acquisition conditions between the two algorithms artificially raise the performance upper bound of the MPC baseline. In practical engineering scenarios, accurate full prediction of future tasks is hardly achievable. It is necessary to highlight the information asymmetry between the compared algorithms in the experimental illustration. Additionally, MPC with imperfect prediction (i.e., prediction noise) can be added as a supplementary baseline. If no new experiments are supplemented, the discussion section should analyze the performance degradation of MPC and the corresponding changes in the advantages of DQN family algorithms when future task prediction errors exist.*

Response. We now state this asymmetry in Methods, Results captions, and Discussion. The new greedy rollout deliberately receives the same forecast as MPC. We did not add a noisy-MPC experiment because a single arbitrary noise distribution would not represent operational forecast quality. Instead, we discuss calibrated prediction distributions and structural link-state errors as necessary future tests. The present comparison is conservative with respect to DQN because only the planners receive future tasks and link states.

Revised manuscript text.

Both MPC and greedy rollout receive perfect next-task and link-state previews; learned policies receive only the current state, so the comparison includes an explicit information asymmetry.

Location in the revised manuscript PDF. pp. 12–13, Section 3.4.4; p. 16, Figure 5 caption; p. 21, Section 4.6; p. 22, Discussion.

Response to Reviewer 3

Comment. *Why was MPC-4 selected as the primary comparison baseline? In addition, the inference time of the DQN models and the decision time of the MPC method should be evaluated and compared on the same hardware platform under the same experimental conditions.*

Response. We now explain that four steps provide nontrivial look-ahead while retaining transparent exhaustive enumeration at every decision. We do not treat $H = 4$ as globally optimal and retain the $H \in \{1, 2, 4, 6\}$ planning-depth trade-off. We added an eight-step greedy rollout and a matched CPU benchmark for all six DQN objectives, MPC-4, and greedy rollout. Measurements use one CPU thread, batch size one, shared saved states, warm-up, and three timed repetitions, and report mean, median, and P95 decision time. The matched comparison appears on pp. 20–21 (Table 12).

Revised manuscript text.

Mean DQN decision time was 0.050–0.051 ms, compared with 2.538 ms for greedy rollout and 4.772 ms for MPC-4.

Location in the revised manuscript PDF. pp. 12–13, Section 3.4.4; p. 18, Table 10; pp. 20–21, Section 4.6 and Table 12.

Comment. *Both the training and evaluation data used in this study are generated by a synthetic simulator, and the reported findings have not yet been validated using real satellite mission data. Although the authors describe the implemented parameter ranges and provide general engineering references for the synthetic simulator, the relationship between its parameter settings and actual satellite operating conditions remains insufficiently explained. The authors are encouraged to clarify how these parameter ranges correspond to representative satellite platforms or mission scenarios and to discuss how potential discrepancies may affect the reported findings. If feasible, validation using real mission data would further strengthen the practical relevance and generalizability of the study.*

Response. We expanded the parameter-provenance explanation and distinguish three categories. Link rates and computing power are engineering-scale anchors supported by cited public sources. Data-reduction ratios and task correlations are synthetic assumptions. Heat, energy, queue, utilization, bandwidth, and contact coordinates are normalized accounting states rather than a calibrated spacecraft digital twin. The Discussion now explains that mission telemetry and hardware-in-the-loop validation are required before operational transfer.

Revised manuscript text.

The parameters serve three distinct roles. Compute power and link rates are engineering-scale anchors supported by public NASA ranges, including SmallSat computing-system power and ground-network examples from 2.2 Mbit s⁻¹ S-band to 220 Mbit s⁻¹ X-band returns [?, ?]. Data-reduction factors and task correlations are synthetic workload assumptions; the reduction factors represent the bandwidth-reduction role of onboard image compression rather than an instrument-specific codec [?]. Heat, energy, queue, utilization, bandwidth, and contact coordinates are normalized accounting states rather than calibrated hardware variables. Table ?? therefore defines a transparent engineering envelope, not a fitted flight distribution or spacecraft digital twin; mission-specific discrepancies in these scales and joint distributions may change absolute costs and potentially the method ordering.

Location in the revised manuscript PDF. p. 6, Section 3.1.2; p. 7, Table 3; p. 23, Sections 5.1–5.2; p. 24, Conclusions.

Reproducibility and availability

The public package contains 1,020 checkpoints and corresponding training curves, locked raw evaluations, seed-level summaries, bootstrap and Holm-adjusted inference, code, validation manifests, and SHA-256 checksums. The exact code-and-data version used for this revision is commit 3076c79 (the full SHA is reported in the manuscript) at https://github.com/noob32123/dqn_family_satellite_ground. The package verifier and all 28 unit tests pass.

We appreciate the opportunity to revise the manuscript and hope that the added experiments, narrower claims, and explicit evidence boundaries address the reviewers’ concerns.

Sincerely,

Bocheng Yang, on behalf of all authors